



Myocardial Infarction

What is Myocardial Infarction (MI)?

- ▶ MI is the death of heart muscle tissue due to prolonged ischemia (lack of oxygen).
- ▶ Caused mainly by coronary artery blockage from plaque rupture and thrombus formation.
- ▶ "Time is muscle": rapid intervention saves heart tissue and lives.

Pathophysiology: How MI Happens

- ▶ Coronary arteries supply oxygen-rich blood to heart muscle; blockage causes ischemia.
- ▶ Plaque rupture exposes thrombogenic material → rapid clot formation → artery occlusion.
- ▶ Most MIs involve the left ventricle, especially the Left Anterior Descending artery (LAD).

Causes

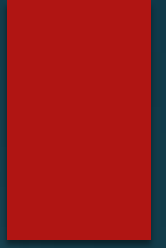
- ▶ Atherosclerosis (most common):
plaque rupture → thrombosis →
occlusion
- ▶ Coronary artery spasm
- ▶ Emboli
- ▶ Severe anemia or hypoxia (rare)

B. Risk Factors

- ▶ Modifiable: smoking, hypertension, diabetes, high cholesterol, obesity, sedentary lifestyle, stress
- ▶ Non-modifiable: age, male sex, family history, genetic predisposition

Types of Myocardial Infarction

STEMI (ST-Elevation MI)



- ▶ **Complete occlusion of a coronary artery**
- ▶ **ST elevation on ECG**
- ▶ **Requires immediate reperfusion (PCI or thrombolytics)**

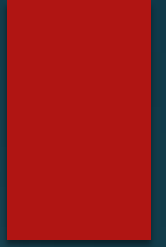
NSTEMI (Non-ST-Elevation MI):

- ▶ Partial occlusion
- ▶ No ST elevation; elevated cardiac biomarkers
- ▶ Managed with medication + possible PCI

Classic Signs & Symptoms

- ▶ Crushing chest pain, often radiating to left arm, jaw, neck, or back.
- ▶ Shortness of breath, nausea, cold sweat, anxiety, and palpitations.
- ▶ Some patients (especially women, elderly, diabetics) may have atypical or silent symptoms.

Diagnostic Tools for MI



- ▶ Electrocardiogram (ECG): ST elevation (STEMI) or ST depression/T wave inversion (NSTEMI).
- ▶ Cardiac biomarkers: Troponin rise confirms myocardial injury.
- ▶ Additional tests: Echocardiogram, stress tests, coronary angiography.

Immediate Nursing Interventions:

- ▶ Morphine: pain relief and anxiety reduction.
- ▶ Oxygen: improve oxygen supply if hypoxic.
- ▶ Nitrates: dilate coronary arteries to improve blood flow.
- ▶ Aspirin: prevent further clotting.
- ▶ Establish IV access and continuous cardiac monitoring.

Advanced Treatments & Procedures

- ▶ Percutaneous Coronary Intervention (PCI): balloon angioplasty and stent placement.
- ▶ Thrombolytic therapy: clot-busting drugs if PCI unavailable.
- ▶ Coronary Artery Bypass Grafting (CABG) for severe blockages.
- ▶ Goal: reperfusion within 90 minutes of hospital arrival.

Complications of MI

- ▶ Arrhythmias (ventricular fibrillation, ventricular tachycardia)
- ▶ Heart failure
- ▶ Cardiogenic shock
- ▶ Papillary muscle rupture
- ▶ Ventricular aneurysm
- ▶ Pericarditis
- ▶ Sudden cardiac death

Nursing Assessment & Monitoring

- ▶ Frequent vital signs and cardiac rhythm monitoring.
- ▶ Assess pain characteristics and response to treatment.
- ▶ Watch for complications: arrhythmias, cardiogenic shock, heart failure.
- ▶ Emotional support and anxiety management.

Post-MI Care & Patient Education

- ▶ Medication adherence: beta blockers, ACE inhibitors, statins, antiplatelets.
- ▶ Lifestyle changes: smoking cessation, diet, exercise, weight control.
- ▶ Recognize warning signs of recurrent MI.
- ▶ Referral to cardiac rehabilitation programs.

Summary & Key Takeaways

- ▶ MI is a life-threatening emergency requiring rapid recognition and treatment.
- ▶ Nurses play a critical role in early assessment, intervention, and education.
- ▶ Understanding MI pathophysiology and care protocols improves patient outcomes.
- ▶ Remember: timely action saves heart muscle and lives.

THANK YOU

